

Question 1

1.1 B✓✓

1.2 B✓✓

1.3 B✓✓

1.4 C✓✓

1.5 D✓✓

1.6 B✓✓

1.7 C✓✓

1.8 A✓✓

1.9 C✓✓

1.10 D✓✓

Question 2

$$\begin{aligned} 2.1 \quad F_n &= mg + F_v \\ &= 40(9,8) + 200 \sin 30 \\ &= 492 \text{ N} \end{aligned}$$

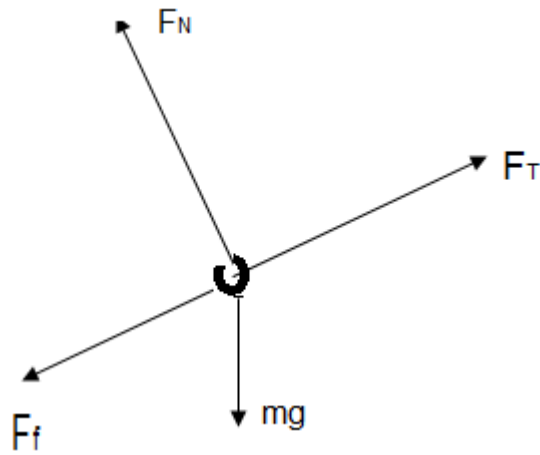
$$\begin{aligned} 2.2 \quad f_{s \max} &= \mu F_N \\ &= (0,4)(492) = 196 \text{ N} \end{aligned}$$

$$\begin{aligned} 2.3 \quad f_k &< F_{s \max} \\ 200 \cos 30 &< 196 \end{aligned}$$

Force applied is smaller than friction force

Child's foot will not slip

QUESTION 3



3.1

3.2 $F_N = F_g \cos \theta$

$$= 2(9,8) \cos 20 = 18,42 \text{ N}$$

3.3 $f_f = \mu F_N = (0,4)(18,42) = 7,37 \text{ N}$

3.4 2 kg

3 kg

$$F = ma$$

$$F = ma$$

$$F_T - f_f - mg \sin \theta = ma$$

$$mg - F_T = ma$$

$$F_T - 7,37 - 2(9,8) \sin 20 = 2a \quad (1) \quad 3(9,8) - F_T = 3a \quad (2)$$

$$F_T = F_T$$

$$7,37 + 2(9,8) \sin 20 + 2a = 3(9,8) - 3a$$

$$a = 3,07$$

$$F_T = -3(3,07) + 3(9,8)$$

$$= 20,19 \text{ N}$$

3.5 $E_{\text{lost}} = W = f_f \Delta x \cos \theta$

$$7,37 \times 0,3 \cos 180 = -2,21 \text{ J}$$

3.6 $mg \sin \theta + f_f = ma$

$$2(9,8) \sin 20 + 7,37 = 2a \quad a = 7,04 \text{ m.s}^{-2}$$

QUESTION 4

4.1 $v_f^2 = v_i^2 + 2 a \Delta y$ ✓

$6^2 = (0)^2 + 2(9,8)\Delta y$ ✓

Up as -

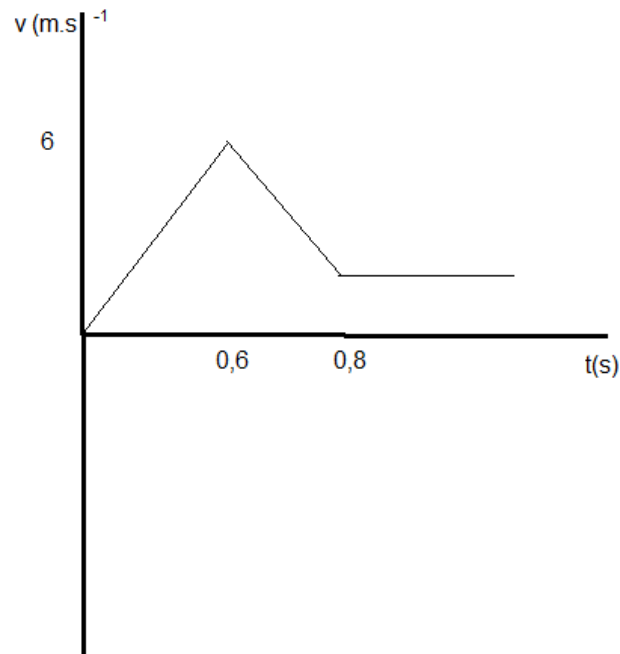
$\Delta y = 1,832 = 3 \text{ m.s}^{-1}$ ✓

(3)

4.2 $v_f = v_i + a \Delta t$

$6 = 0 + 9,8 \Delta t$ $\Delta t = 0,61 \text{ s}$

4.3



4.4 Impulse is the product of a force applied on an object and time it is applied

4.5 $F \Delta t = m(v_f - v_i)$ ✓

$F(0,2) = 0,3(0,8 - 6)$ ✓

$F = 7,85 \text{ N upwards}$ ✓

is +
↓

4.6 $F_{\text{net}} = F_{\text{applied}} - mg$

$$-7,8 = F_{\text{applied}} + 0,3(9,8)$$

$$F_{\text{applied}} = 10,72 \text{ N upwards}$$

- 4.7 gravitational force is equal to the upwards of the water, therefore the ball has not net force and moves at a constant velocity

QUESTION 5

- 5.1 The net work done by a net force is equal to the change in kinetic energy of the system ✓✓

$$5.2 \text{ Gradient} = \frac{\Delta y}{\Delta x} = \frac{\Delta Ek}{\Delta x}$$

$$\text{Formula } W = F_{\text{net}} \Delta x \cos \theta \quad \text{Thus } F_{\text{net}} = \frac{W}{\Delta x \cos \theta} = \frac{\Delta Ek}{\Delta x \cos \theta}$$

$$\text{Gradient} = F_{\text{net}}$$

$$5.3 \quad F_{\text{net(c)}} = \text{Gradient} = \frac{\Delta y}{\Delta x} = \frac{180-88}{15-10} = 18,4 \text{ N}$$

$$5.4 \quad F_{\text{net}} = F_{\text{app}} - f_f$$

$$18,4 = 20 - f_f$$

$$F_f = 1,6 \text{ N}$$

- 5.5 Larger friction force, smaller net force as applied force stays constant.

smaller net force smaller gradient

B has the smallest gradient

- 5.7 power is the rate of doing work

$$5.8 \quad P = F \times v$$

$$60\,000 = F \times \frac{60}{3,6}$$

$$F = 3\,600 \text{ J}$$

QUESTION 6

6.1 Linear momentum in an isolated system is conserved ✓✓ (2)

6.2 $\Sigma p_{\text{initial}} = \Sigma p_{\text{final}}$

$$m_1 v_1 + m_2 v_2 = (m_1 + m_2) v_f \quad \checkmark$$

$$1500 \left(\frac{70}{3,6} \right) + m(0) = (1500 + m)(7) \checkmark$$

$$m = 2666,67 \text{ m.s}^{-1} \checkmark$$

6.3 $E_k \text{ before} = \frac{1}{2} m v^2$

$$\frac{1}{2} \times 1500 \times \left(\frac{70}{3,6} \right)^2$$

$$= 283\,564 \text{ J}$$

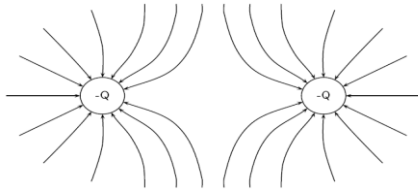
Kinetic energy is not conserved

$$E_k \text{ after} = \frac{1}{2} m v^2$$

$$\frac{1}{2} \times (1500 + 2666,67) \times (7)^2$$

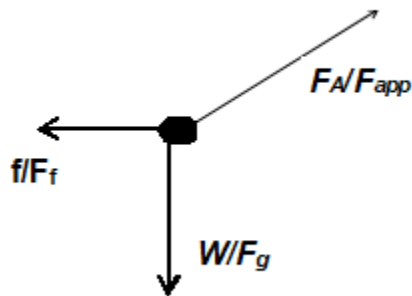
$$= 102083,33 \text{ J}$$

QUESTION 7



7.1.

7.2 $w = mg = 0,002 \times 9,8 = 0,0196 \text{ N down}$



7.3

7.4 The magnitude of the electrostatic force exerted by one point charge (Q_1) on another point charge (Q_2) is directly proportional to the product of the magnitudes of the charges and inversely proportional to the square of the distance (r) between them. ✓✓ (2)

7.5

$$F_T \sin \theta = mg$$

$$F_t = \frac{0,0196}{\sin 70} = 0,0208 \text{ N}$$

$$F = \frac{kQ_1q_2}{r^2} \quad \checkmark$$

$$0,0208 \checkmark = \frac{9 \times 10^9 q^2}{0,01^2} \quad \checkmark$$

$$q = 8,89 \times 10^{-9} \text{ C} \quad \checkmark$$

7.6 $E = \frac{kq}{r^2}$

$$\frac{9 \times 10^9 8,89 \times 10^{-9}}{0,3^2}$$

$$E = =$$

7.7.1 The attraction force exerted on any two objects in the universe are directly proportional to the product of the magnitudes of their masses and inversely proportional to the square of the distance (r) between their centres. ✓✓ (2)

7.7.2

$$F(\text{earth}) = 70(9,8) = 686 \text{ N}$$

$$F_{\text{earth}} = \frac{kM_1M_2}{r^2} \checkmark$$

$$F_{\text{Beeble}} = \frac{k \frac{1}{2} \checkmark M_1M_2}{(\frac{1}{4} \checkmark r)^2} = F_{\text{Beeble}} = \frac{8 \times kM_1M_2}{r^2}$$

$$F_{\text{Beeble}} = 8 \times 686 \checkmark = 5488 \text{ N } \checkmark \quad (5)$$

QUESTION 8

$$8.1 \ P = I^2 R = 2,31^2 \times 10 = 53,36 \text{ W}$$

$$\begin{aligned} 8.2. \quad EMF &= I(R + r) \\ 30 &= 2,31(R + 1) \\ R &= 11,99 \ \Omega \end{aligned}$$

(3)

$$8.3 \quad 0 \text{ V}$$

$$\begin{aligned} 8.4 \quad V &= IR && \checkmark \\ V &= 2,31 \times 11,99 && \checkmark \\ v &= 27,69 \text{ V} && \checkmark \end{aligned}$$

8.5 *Resistance is the opposition to the flow of the current*

8.6.1 DECREASE

8.6.2 As resistance decrease in parallel, current increase in series circuit as resistance is inversely proportional to current

Power in motor increase as $P = I^2 R$ and the resistance of motor stays the same

$$8.7.1 \quad \frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{1}{R_T} = \frac{1}{5} + \frac{1}{10}$$

$$R_t = 3,33 \Omega$$

$$Emf = IR + Ir \quad \checkmark$$

$$30 = 13,33 + 1 \quad \checkmark$$

$$I = 2,09 \text{ A} \quad \checkmark$$

$$8.7.2 \quad V = IR \quad \checkmark$$

$$V = 3,33 \times 2,09 \quad \checkmark$$

$$V = 6,97 \text{ V}$$